



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (CSE/ISE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CS303 / 20CS304 / 21IS303 / 20IS304 - COMPUTER ORGANIZATION AND ARCHITECTURE

19IS304 - COMPUTER ORGANIZATION

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	With a neat diagram explain the functional units of a computer?	5	L*2	1	1
	b)	With a neat diagram showing the connection between processor and memory, list the steps for the execution of any instruction.	8	L2	1	1
	c)	List and explain all the factors which play a key role in improving the performance of the computer.	7	L2	1	1
2.	a)	Highlight the difference between primary and secondary memory.	6	L2	1	1
	b)	Define addressing mode. List and explain any four addressing modes with example.	8	L2	1	1
	c)	With a neat diagram explain the working of sequential multiplication technique for solving the 1100 x1001.	6	L3	2	1
3.	a)	Explain the steps for Booth multiplication for multiplying (-17) x(-15).	5	L3	2	1
	b)	Explain how integer division takes place in computer with a neat circuit arrangement.	5	L2	2	1
	c)	Divide 18/5 using the restoring division technique.	5	L3	2	2
	d)	Explain the IEEE standard for single precision floating-point numbers.	5	L2	2	1
		Unit – II				
4.	a)	With a neat diagram, explain the working of data path of single bus organization processor.	7	L2	3	1
	b)	Write the control sequence for the execution of conditional branch instruction in a single-bus organization.	7	L3	3	1
	c)	Write the control sequence and microinstruction for execution of ADD (R3), R1 instruction on a 3-bus architecture.	6	L3	3	1
5.	a)	With block diagram, explain the basic organization of a microprogrammed control unit.	6	L2	3	1
	b)	With a neat diagram explain the organization of a 2Mx8 memory chip.	6	L2	4	2
	c)	Explain synchronous DRAM with neat diagram and also illustrate its working for burst reading of length 4 with timing signals.	8	L2	4	1
6.	a)	What is the need of ROM? List and explain its types.	5	L1	4	1
	b)	Define the mapping function. Explain different type of mapping function with example.	10	L2	4	1
	c)	What is virtual memory? Explain address translation with a neat diagram.	5	L2	4	1
		Unit – III				
7.	a)	Explain the timing diagram for synchronous and asynchronous buses.	8	L2	5	1
	b)	What is interrupt? Explain enabling and disabling interrupts.	6	L2	5	1
	c)	Explain with a neat diagram interrupt priority schemes.	6	L2	5	1

P.T.O.

8. a) What is DMA? Explain the use of DM controller in a computer with neat diagram along with registered in DMA interface. 7 L2 5 1
- b) Define bus arbitration. Explain the different types of bus arbitration used in computer with neat diagrams. 7 L2 5 1
- c) What are the different data transfer signals available in PCI bus, explain with a read operation on the PCI bus timing signals.? 6 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
